

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

103000657 - Correctness By Construction

DEGREE PROGRAMME

10AM - Master Universitario En Ingenieria Del Software

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

Index

Learning guide

1. Description.....	1
2. Faculty.....	1
3. Prior knowledge recommended to take the subject.....	2
4. Skills and learning outcomes	3
5. Brief description of the subject and syllabus.....	4
6. Schedule.....	6
7. Activities and assessment criteria.....	9
8. Teaching resources.....	10
9. Other information.....	11

1. Description

1.1. Subject details

Name of the subject	103000657 - Correctness By Construction
No of credits	6 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 2
Tuition period	February-June
Tuition languages	English
Degree programme	10AM - Master Universitario en Ingeniería del Software
Centre	10 - E.T.S. De Ingenieros Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Manuel Carro Liñares (Subject coordinator)	2303	manuel.carro@upm.es	M - 13:00 - 15:00 Tu - 13:00 - 15:00 W - 13:00 - 15:00 Please note that the office hours may change during the course. Please get in touch with the instructor to get an appointment.

Manuel De Hermenegildo Salinas	2212	manuel.hermenegildo@upm. es	Sin horario. Please get in touch with the instructor to get an appointment.
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* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Declarative programming
- First-order logic
- Programming experience (minimum 2 years)
- Formal proofs
- Reasoning about properties of algorithms
- Concurrent programming and concurrent systems

4. Skills and learning outcomes *

4.1. Skills to be learned

CE13 - Tener una visión de los distintos aspectos específicos y emergentes de la ingeniería del software, y profundizar en algunos de ellos

CE14 - Comprender lo que pueden y no pueden conseguir las prácticas actuales de ingeniería del software, y sus limitaciones y su posible futura evolución.

CG13 - Apreciación de los límites del conocimiento actual y de la aplicación práctica de la tecnología más reciente

4.2. Learning outcomes

RA66 - RA-AV-2 Acquaintance with various techniques for formal software development

RA25 - Communication skills in public SC13, SC14, CG3, CG18 S

RA50 - Posee las técnicas necesarias para la realización de un informe o memoria sobre un trabajo realizado en un entorno socio?lingüístico nacional/internacional.

RA69 - RA-AV-4 - Knowledge of techniques for formally proving code correctness.

RA65 - RA-AV-1 - Acquaintance with design requirements and implementation requirements.

RA54 - Proponer una solución justificada a un problema real que sea complejo o mal definido, o perteneciente a un área nueva o emergente, o que requiera el desarrollo de enfoques o métodos nuevos y originales, dentro del contexto de la ingeniería del software justificándola de una forma cualitativa y cuantitativa.

RA26 - Group work skill SC13, SC14, CG17 A

RA68 - RA-AV-3 - Knowledge of languages for formal specification

RA91 - Apply techniques for modelling the context of use

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Software is becoming increasingly complex and responsible for critical tasks. Any technology aimed at ensuring the reliability and quality of software will be increasingly relevant, if not utterly necessary.

Only rigorous (e.g., mathematically sound) approaches can certify software with the highest possible assurance. These approaches include, among others, the use of specification languages, high-level programming languages (including equational, functional, and logic languages), the use of model checking and deductive verification, language-based approaches often interacting with theorem provers.

In this course we will give a hands-on introduction to rigorous software development methods that follow a *correctness-by-construction* approach. While the course is not heavy in theory, everyone is expected to have a good understanding of first-order logic and programming experience.

5.2. Syllabus

1. Introduction to Formal Methods: Proving Programs Correct
2. Fundamentals of Formal Methods: Specification, First-Order Logic, Proofs, Programs
3. Event-B Basics and the Rodin Tool
4. Sequential Systems
5. Event B: Mathematical Toolkit and Applications
6. Reactive Systems: Concurrency and Distribution

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Introduction to formal methods and correctness by construction Duration: 01:30 Lecture Sample cases of formal development Duration: 01:30 Cooperative activities			
2	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
3	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			Homework Individual work Progressive assessment Not Presential Duration: 04:00
4	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
5	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
6	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			Homework Individual work Progressive assessment Not Presential Duration: 04:00

7	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
8	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
9	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			Homework Individual work Progressive assessment Not Presential Duration: 08:00
10	Quizzes Duration: 01:00 Problem-solving class Event-B and related topics Duration: 02:00 Lecture			
11	Presentation of term project Duration: 01:00 Additional activities Event-B and related topics Duration: 02:00 Lecture			Term project Group work Progressive assessment Not Presential Duration: 20:00
12	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
13	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			
14	Event-B and related topics Duration: 02:00 Lecture Quizzes Duration: 01:00 Problem-solving class			

15	Presentaciones de trabajo en grupo Duration: 03:00 Additional activities			Presentation and defense of group projects Group presentation Progressive assessment Presential Duration: 03:00
16				
17				Final regular exam Written test Global examination Presential Duration: 03:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
3	Homework	Individual work	No Presential	04:00	20%	2 / 10	CE13 CE14 CG13
6	Homework	Individual work	No Presential	04:00	20%	2 / 10	CE13 CE14 CG13
9	Homework	Individual work	No Presential	08:00	20%	2 / 10	CE13 CE14 CG13
11	Term project	Group work	No Presential	20:00	%	4 / 10	CE13 CE14 CG13
15	Presentation and defense of group projects	Group presentation	Face-to-face	03:00	40%	4 / 10	CE13 CE14 CG13

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Final regular exam	Written test	Face-to-face	03:00	100%	5 / 10	CE13 CE14 CG13

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Extra final exam	Written test	Face-to-face	03:00	100%	5 / 10	CE13 CE14 CG13

7.2. Assessment criteria

- No mandatory activities are necessary to pass via the final exams
- The minimum grade to pass the course is 5 over 10 (either when it is calculated as the weighted sum of individual homework or when it is the grade of a single comprehensive exam).
- The topics covered in the different homework assignments cannot be tested separately in the final exam, as they are deeply intertwined and are not isolated units of knowledge.
- The global exams, both the regular and the extraordinary ones, will be in person.
- Copying from any source (either textbooks, the Internet, another student, or any other source) with or without the permission of the author of the source, as well as other types of academic fraud, can lead to a 'fail' grade in the course and / or being reported to the academic authorities, who will decide whether to take additional authoritative measures. In particular, in case of non-ethical or fraudulent behavior, the Law 3/2022 of February 24th will be applied, as well as the corresponding UPM regulations. Article 12 and 14 of Law 3/2022 states that a serious fault may mean, among other outcomes, failing the corresponding sitting.
- There are no learning blocks whose earned grades can be carried over to future academic courses.
- Failure to deliver a homework assignment at the time and in the form stated by the instructor(s) may result in a failure for that exercise.
- For progressive evaluation: if for any reason it is not possible to prepare / hand out some homework assignment, its weight in the final grade will be split among the rest of the homework exercises in such a way that the relative weight of the rest of the assignments, when compared with each other, will be the same they had before removing the homework that could not be handed out.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Lawrence Paulson's class notes	Bibliography	Lawrence Paulson's Logic and Proof are the course notes of the author for a Logic course in Cambridge. Highly recommended, as they are both rigorous and very concise. They provide very good background material for both parts of the course.

Logic in Computer Science (Huth and Ryan)	Bibliography	A very good book on the use of logic in computer science is Logic in Computer Science, by Huth and Ryan. The Computer Science School should have several copies. There may be electronic copies on the Internet, if possible of the second edition.
http://wiki.event-b.org/	Web resource	Central Event-B site
Modeling in Event-B: System and Software Engineering, by Jean-Raymond Abrial.	Bibliography	The reference book for Event B, with plenty of worked examples.

9. Other information

9.1. Other information about the subject

This course will be given in English. Please note that in case Spanish appears as the course language in the general description, that would be a clerical mistake.